Parametric Equations

Starter

Solution 2

1. Make t the subject:

a. $x = 4t - kt$

b. $y = 3t^2$

c. $y = 2 - 4\ln t$

d. $x = 1 + 2e^{-3t}$

1

$t = \dfrac{x}{4-k}$; $\quad t = \pm\sqrt{y/3}$

$t = e^{(2-y)/4}$; $\quad t = -\frac{1}{3}\ln((x-1)/2)$

2. Express using cosine:

a. $4 + 3\sin^2 x$

b. $\sin 2x$

c. $\cot x$

d. $2\cos x + \cos 2x$

2

$7 - 3\cos^2 x$; $\quad \pm 2\cos x\sqrt{1-\cos^2 x}$

$\dfrac{\pm\cos x}{\sqrt{1-\cos^2 x}}$; $\quad 2\cos^2 x + 2\cos x - 1$

The signs depend on $\sin x$; cotangent is undefined when $\sin x = 0$.

3. Find the ranges:

a. $\ln(x+1),\ x > 0$

b. $2\sin x,\ 0 < x < \pi$

c. $x^2 + 4x - 2,\ -4 < x < 1$

d. $1/(2x+5),\ x > -2$

a

a. $y > 0$

b. $0 < y \leq 2$

c. $-6 \leq y < 3$

d. $0 < y < 1$

4. Find the intersections of the circle with centre $(0, 4)$, radius 5, and the line $2y - x - 10 = 0$.

4

$x^2 + (x/2+1)^2 = 25$

$(x, y) = (4, 7),\ (-24/5, 13/5)$

Chapter Overview

Solution 3

Chapter overview

1. Convert $x = 2\cos t + 1,\ y = 3\sin t$ to Cartesian form.
2. Sketch $x = 2t,\ y = 5/t$.
3. Find where $x = t^2,\ y = 4t$ meets $x + y + 4 = 0$.
4. Model a plane using $x = (v\cos\theta)t,\ y = (v\sin\theta)t$.

What are they and what is the point?

Solution 4

A Cartesian equation relates x and y, for example $x^2 - y^2 = 5$.

Parametric equations express both coordinates using another variable, such as $x = \sin t,\ y = t^2$.

Each allowed value of t gives a point on the curve. Eliminating t does not always preserve the full domain or direction.

Converting parametric to Cartesian

Solution 5

Find a Cartesian equation and the domain and range of

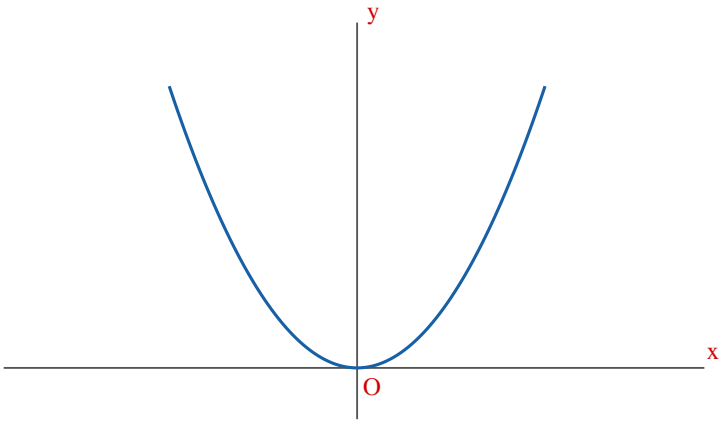
$x = 2t, \quad y = t^2, \quad -3 < t < 3.$

Answer

$t = x/2 \Rightarrow y = x^2/4$

$-6 < x < 6, \qquad 0 \leq y < 9$

Sketch



Further Example

Solution 6

Find a Cartesian equation and state its domain and range:

$x = \ln(t+3), \quad y = 1/(t+5), \quad t > -2.$

Answer

$t = e^x - 3 \Rightarrow y = \dfrac{1}{e^x + 2}$

$x > 0, \quad 0 < y < \frac{1}{3}$

Now you try.

Solution 7

1. Eliminate t: $x = \ln(t+2),\ y = 1/(t+1),\ t > -1$.

1

$t = e^x - 2 \Rightarrow y = \dfrac{1}{e^x - 1}, \quad x > 0$

2. Eliminate t: $x = \ln t,\ y = t^2 - 2,\ t > 0$.

2

$t = e^x \Rightarrow y = e^{2x} - 2, \quad x \in \mathbb{R}$

Pure Year 2/AS Exercise 8A (200-202)

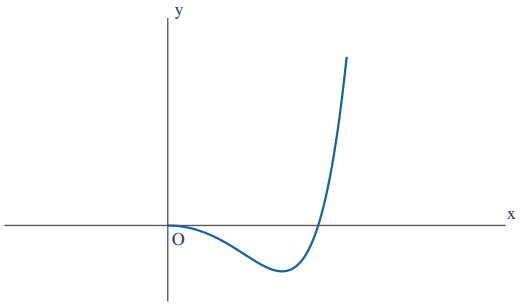
Solution 8

8. A curve is given by $x = 3\sqrt{t},\ y = t^3 - 2t,\ 0 \le t \le 2$.

a. Find its Cartesian equation and domain.

b. Find where $dy/dt = 0$.

c. Hence find the range of y.



a

$t = x^2/9 \Rightarrow y = \dfrac{x^6}{729} - \dfrac{2x^2}{9}$

$0 \le x \le 3\sqrt{2}$

b

$3t^2 - 2 = 0 \Rightarrow t = \sqrt{2/3}$

c

$-\dfrac{4\sqrt{6}}{9} \le y \le 4$

Compare the stationary value with both endpoints.

Pure Year 2/AS Exercise 8A (200-202) — Solution 9

9. $x = t^3 - t,\ y = 4 - t^2,\ t \in \mathbb{R}$.

a. Show that $x^2 = (a - y)(b - y)^2$, finding a, b.

b. Find the greatest value of y.

a

$x = t(t^2 - 1), \quad t^2 = 4 - y$

$x^2 = (4 - y)(3 - y)^2, \quad a = 4,\ b = 3$

b

$y \leq 4$ with equality at $t = 0$

...when you have trig identities — Solution 10

1. Eliminate t from $x = \sin t + 2,\ y = \cos t - 3$.

1

$(x - 2)^2 + (y + 3)^2 = 1$

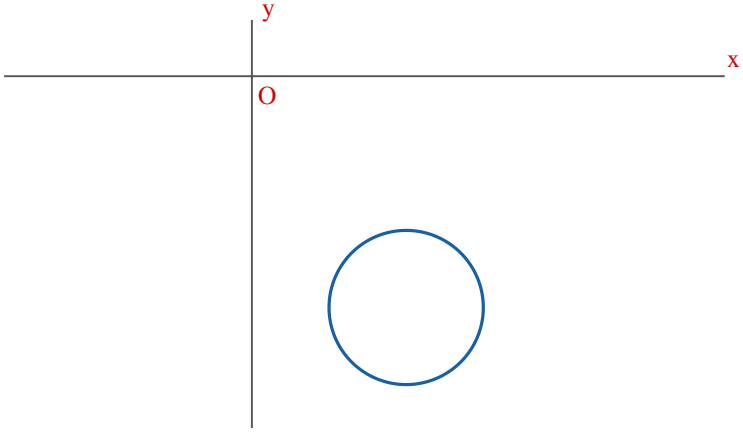
2. Eliminate t from $x = \sin t,\ y = \sin 2t,\ -\pi/2 \leq t \leq \pi/2$.

2

$y = 2x\sqrt{1 - x^2}, \quad -1 \leq x \leq 1$

Here $\cos t \geq 0$, and $-1 \leq y \leq 1$.

1: Sketch of the circle (centre (2, -3), radius 1)



Now you try..

Solution 11

1. $x = 2\sin t,\ y = 1 - \cos 2t,\ -\pi/2 \le t \le \pi/2$. Find a Cartesian equation and its domain.

1 $y = 2\sin^2 t = x^2/2, \quad -2 \le x \le 2$

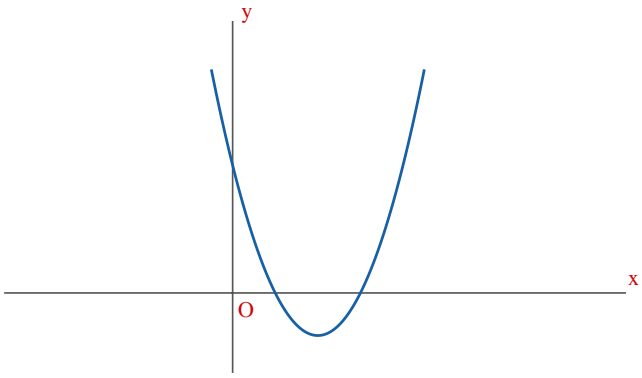
2. $x = \cot t + 2,\ y = \csc^2 t - 2,\ 0 < t < \pi$. Find a Cartesian equation and sketch the curve.

2

$y = (x-2)^2 - 1 = x^2 - 4x + 3$

All real x; vertex $(2, -1)$, roots 1 and 3.

Sketch



Exercise 8B Pure 2 pages 204-206

Solution 12

6. $x = 8\cos t,\ y = \frac{1}{4}\sec^2 t,\ -\pi/2 < t < \pi/2.$

Find a Cartesian equation, its domain and range. The endpoints $t = \pm\pi/2$ are excluded because $\sec t$ is undefined.

Answer

$y = \dfrac{16}{x^2}, \quad 0 < x \le 8, \quad y \ge \dfrac{1}{4}$

Exercise 8B Pure 2 pages 204-206 — Solution 13

7. $x = 3\cot^2 2t,\ y = 3\sin^2 2t,\ 0 < t \le \pi/4$. Find a Cartesian equation and its domain.

Answer

$x = 3(\csc^2 2t - 1) = 9/y - 3$

$y = \dfrac{9}{x+3}, \quad x \ge 0$

Exercise 8B Pure 2 pages 204-206 — Solution 14

8. $x = \frac{1}{3}\sin t,\ y = \sin 3t,\ 0 < t < \pi/2$.

a. Show $y = ax(1 - bx^2)$, finding a, b.

b. State the domain and range.

a

$\sin 3t = 3\sin t - 4\sin^3 t$

$y = 9x - 108x^3 = 9x(1 - 12x^2)$

$a = 9, \quad b = 12$

b

$0 < x < \frac{1}{3}, \qquad -1 < y \le 1$

Exercise 8B Pure 2 pages 204-206 Solution 15

10. $x = \tan^2 t + 5,\ y = 5\sin t,\ 0 < t < \pi/2.$

a. Find a Cartesian equation.

b. State the domain and range.

a

$$y^2 = 25\frac{\tan^2 t}{1+\tan^2 t} = 25\frac{x-5}{x-4}$$

b

$$x > 5, \quad 0 < y < 5$$

Exercise 8B Pure 2 pages 204-206 Solution 16

11. $x = \tan t,\ y = 3\sin(t-\pi),\ 0 < t < \pi/2.$ Find a Cartesian equation and its domain and range.

Answer

$$\sin(t-\pi) = -\sin t = -\frac{\tan t}{\sqrt{1+\tan^2 t}}$$

$$y = -\frac{3x}{\sqrt{1+x^2}}, \quad x > 0, \quad -3 < y < 0$$

Sketching a spiral

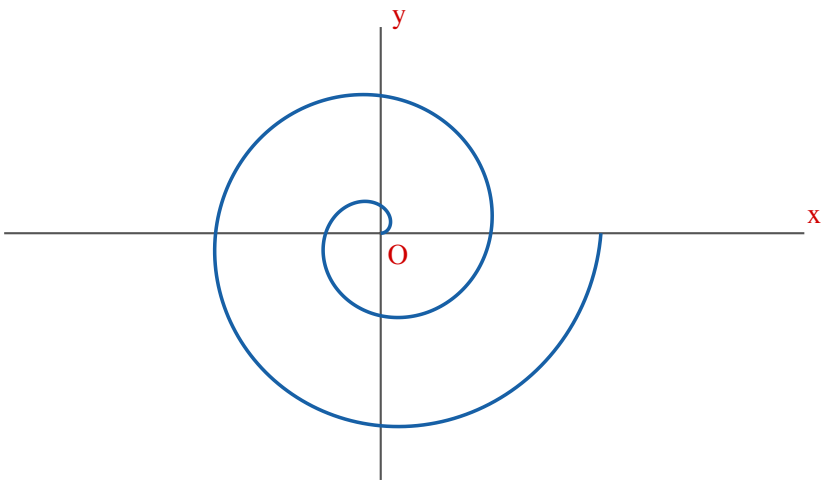
$x = \theta\cos\theta, \quad y = \theta\sin\theta, \quad 0 \le \theta \le 4\pi.$

Mark the points for $\theta = 0, \pi/2, \pi, 3\pi/2, 2\pi$. What happens during the second turn?

Sketch

The distance from the origin is θ. The curve winds anticlockwise outwards, completing two turns. Inverse cosine alone cannot give a single Cartesian description of all turns.

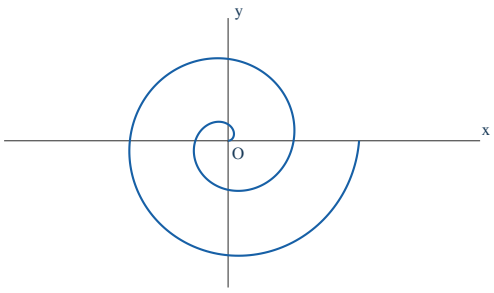
Sketch



Plot points in parameter order

θ	0	$\pi/2$	π	$3\pi/2$	2π
x	0	0	$-\pi$	0	2π
y	0	$\pi/2$	0	$-3\pi/2$	0

For $x = \theta\cos\theta,\ y = \theta\sin\theta$, plot further points up to 4π. Join them smoothly and show the direction as θ increases.



Now you try..

Solution 19

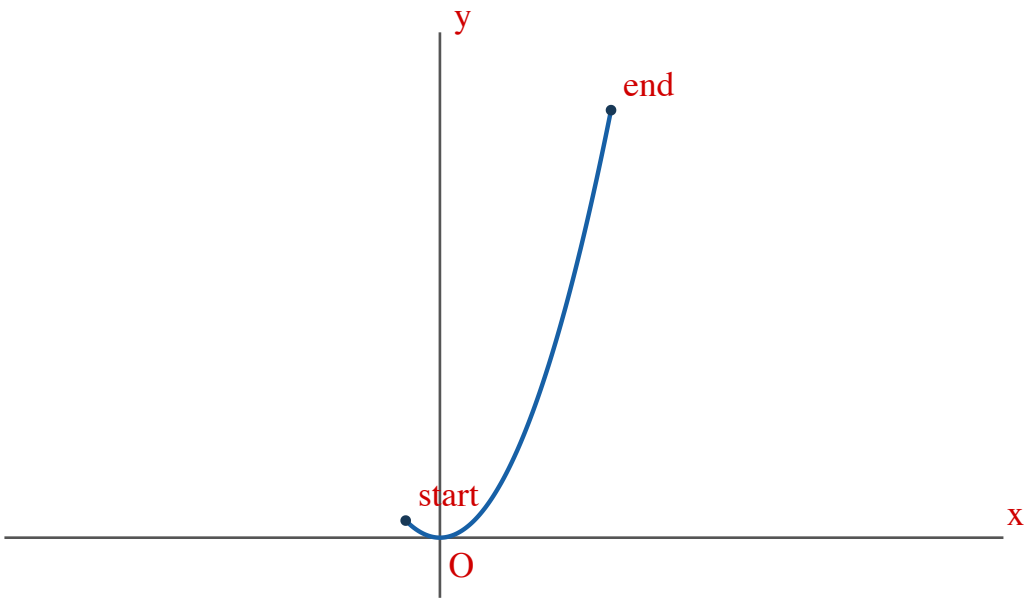
Sketch $x = 2t,\ y = t^2,\ -1 \le t \le 5$, showing endpoints and direction.

Sketch

$y = x^2/4, \quad -2 \le x \le 10$

Start at $(-2, 1)$, pass through $(0, 0)$, and finish at $(10, 25)$.

Sketch



Exercise IM Level 1 Exercise 1

Solution 20

Find Cartesian equations:

i. $x = 1 - t,\ y = t^2 - 4$

ii. $x = 2t^2,\ y = 1/t,\ t \ne 0$

iii. $x = 2\cos\theta + \sin\theta,\ y = \cos\theta - 2\sin\theta$

i

$y = (1 - x)^2 - 4 = x^2 - 2x - 3$

ii

$xy^2 = 2, \quad x > 0,\ y \ne 0$

iii

$x^2 + y^2 = 5(\sin^2\theta + \cos^2\theta) = 5$

Exercise IM Level 1 Exercise 2 — Solution 21

For $x = t^2,\ y = t^3$, complete a table for integer t from -3 to 3, sketch the curve and find a Cartesian equation.

Table

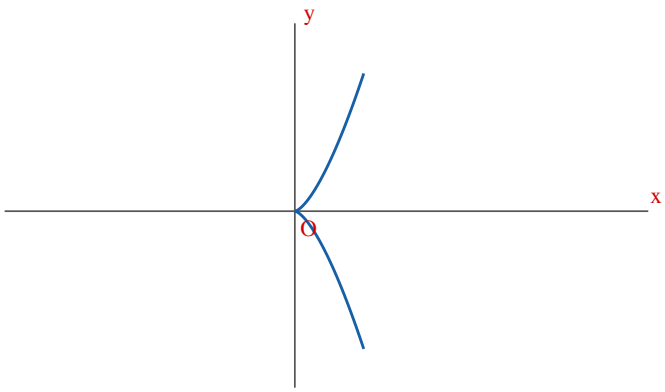
t	-3	-2	-1	0	1	2	3
x	9	4	1	0	1	4	9
y	-27	-8	-1	0	1	8	27

Equation

$y^2 = x^3, \quad x \geq 0$

The curve has two branches meeting in a cusp at the origin.

Sketch



Exercise 8C (Pages 207-208) — Solution 22

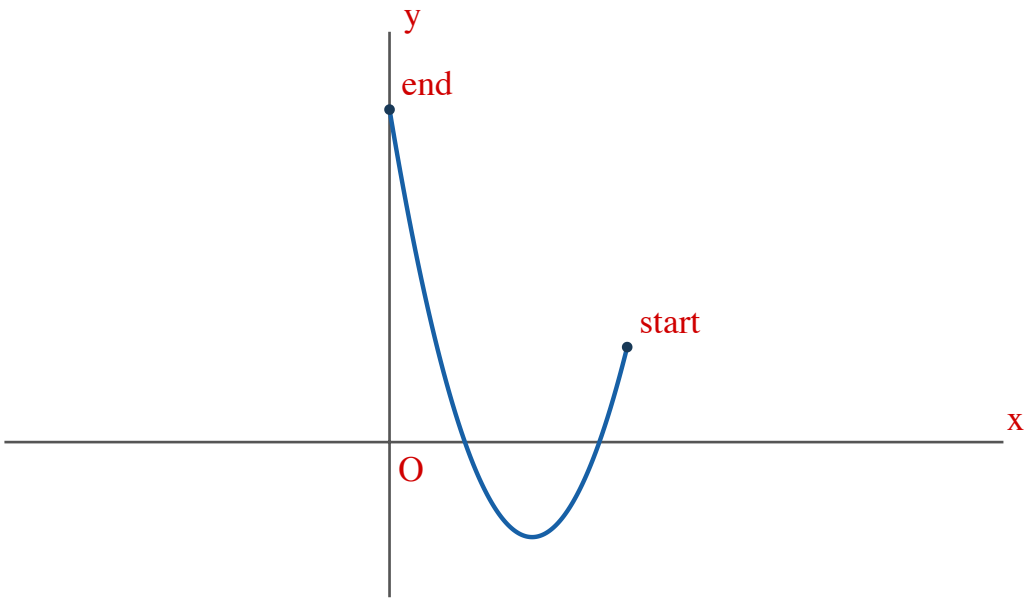
Find the Cartesian equation and sketch $x = 3 - t,\ y = t^2 - 2,\ -2 \leq t \leq 3$.

Answer

$y = (3 - x)^2 - 2, \quad 0 \leq x \leq 5$

The vertex is $(3, -2)$. As t increases the curve runs from $(5, 2)$ to $(0, 7)$.

Sketch



Exercise 8C (Pages 207-208) — Solution 23

$x = 9\cos t - 2,\ y = 9\sin t + 1,\ -\pi/6 \le t \le \pi/2.$

a. Find a Cartesian equation.

b. Sketch the arc, giving its endpoints.

c. Find its length.

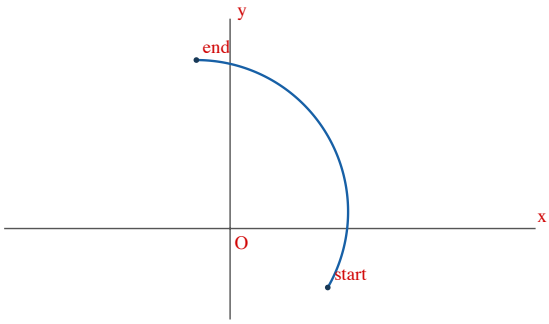
a $(x+2)^2 + (y-1)^2 = 81$

b

Centre $(-2, 1)$, radius 9. Anticlockwise from $(9\sqrt{3}/2 - 2, -7/2)$ through $(7, 1)$ to $(-2, 10)$.

c $s = 9(\pi/2 + \pi/6) = 6\pi$

Sketch



Points of Intersection — Solution 24

$x = at^2 + t,\ y = a(t^3 + 8)$ passes through $(-4, 0)$. Find a and the intersections with the y-axis.

Find a

$t = -2, \quad 4a - 2 = -4 \Rightarrow a = -\frac{1}{2}$

Intersections

$x = -t^2/2 + t = 0 \Rightarrow t = 0, 2$

$(x, y) = (0, -4), (0, -8)$

Points of Intersection

Solution 25

1. Find where $x = t^2,\ y = 4t$ meets $x + y + 4 = 0$.

1

$t^2 + 4t + 4 = 0 \Rightarrow t = -2$

$(x, y) = (4, -8)$

The repeated root indicates tangency.

2. $x = \cos t + \sin t,\ y = (t - \pi/6)^2,\ -\pi/2 < t < 4\pi/3$. Find points with $y = \pi^2$, and intersections with the y-axis.

2: given y

$t = 7\pi/6 \Rightarrow (x, y) = (-(1 + \sqrt{3})/2, \pi^2)$

The other candidate $-5\pi/6$ is outside the domain.

2: axis

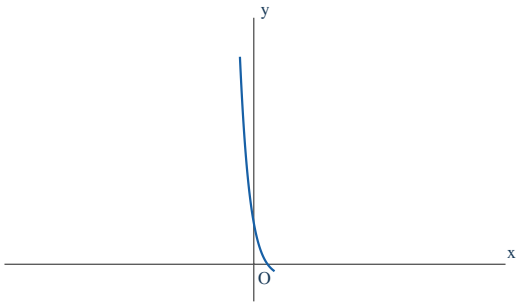
$t = -\pi/4, 3\pi/4 \Rightarrow y = 25\pi^2/144, 49\pi^2/144$

Now you try

Solution 26

C4 January 2013

$x = 1 - t/2,\ y = 2^t - 1$. Find the intercepts with both axes.



y-axis

$x = 0 \Rightarrow t = 2 \Rightarrow (0, 3)$

x-axis

$y = 0 \Rightarrow t = 0 \Rightarrow (1, 0)$

Exercise 8D (page 211-213) Solution 27

3. $x = 4at^2,\ y = a(2t - 1)$ passes through $(4, 0)$. Find a.

Answer

$a \neq 0, \quad t = 1/2, \quad 4 = 4a(1/2)^2 \Rightarrow a = 4$

Exercise 8D (page 211-213) Solution 28

4. $x = b(2t - 3),\ y = b(1 - t^2)$ passes through $(0, -5)$. Find b.

Answer

$b \neq 0, \quad t = 3/2, \quad -5 = b(1 - 9/4) \Rightarrow b = 4$

Exercise 8D (page 211-213) Solution 29

5. Find the intersection of $x = 3t + 2,\ y = 1 - t$ with the line $x + y = 2$.

Answer

$3t + 2 + 1 - t = 2 \Rightarrow t = -1/2$

$(x, y) = (1/2, 3/2)$

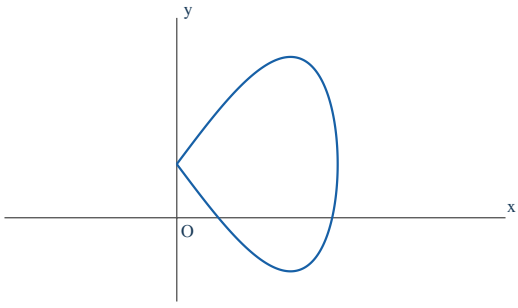
Exercise 8D (page 211-213) Solution 30

12. $x = 6\cos t,\ y = 4\sin 2t + 2,$ $-\pi/2 < t < \pi/2.$

a. Find the x-axis intercepts.

b. Find the parameter values for intersections with $y = 4$.

c. Give the coordinates of these points.



a

$\sin 2t = -1/2 \Rightarrow t = -\pi/12, -5\pi/12$

Intercepts: $(6\cos(\pi/12), 0)$, $(6\cos(5\pi/12), 0)$.

b

$\sin 2t = 1/2 \Rightarrow t = \pi/12, 5\pi/12$

c

$(6\cos(\pi/12), 4)$ and $(6\cos(5\pi/12), 4)$.

Exercise 8D (page 211-213) Q13

Solution 31

Show that the line $y = 2x - 5$ does not intersect $x = 2t,\ y = 4t(t-1)$.

Answer

$4t(t-1) = 4t - 5 \Rightarrow 4t^2 - 8t + 5 = 0$

$\Delta = 64 - 80 = -16 < 0$

There is no real value of t, so no intersection.

Exercise 8D (page 211-213) Q14

Solution 32

$x = \sin t,\ y = \cos 2t + 1,\ 0 \leq t \leq 2\pi$.

a. For which values of k does $y = k$ meet the curve?

b. Find a Cartesian equation and the horizontal tangent.

a

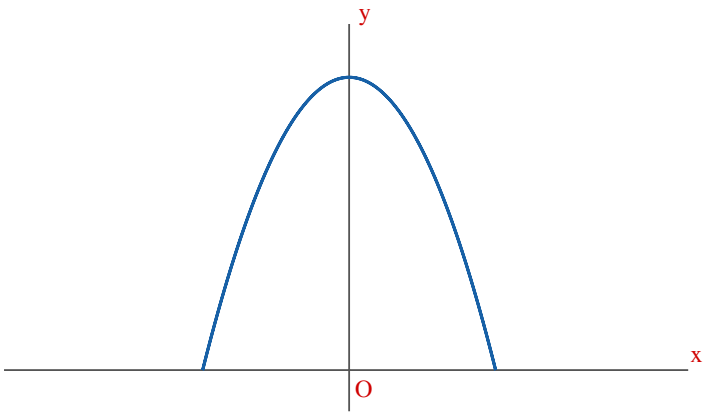
$0 \leq k \leq 2$

b

$y = 2 - 2x^2, \quad -1 \leq x \leq 1$

The horizontal tangent is $y = 2$, at $(0, 2)$.

Sketch



Modelling

Solution 33

A plane travels at constant speed v, making angle θ with the horizontal. It rises 120 m for every 600 m horizontally.

a. Find θ.

b. Give parametric equations when $v = 50\,\text{m s}^{-1}$.

c. Find its height after 10 seconds and a Cartesian equation. Comment on the model.

a

$\theta = \arctan(1/5) \approx 11.3^\circ$

b

$$x = \frac{250}{\sqrt{26}}t, \quad y = \frac{50}{\sqrt{26}}t$$

c

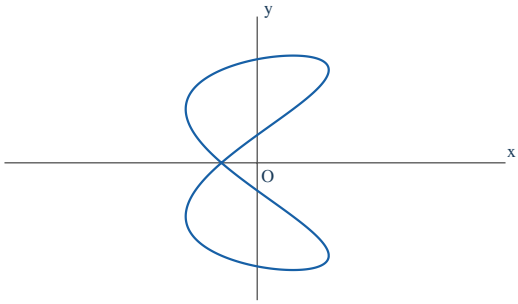
$$y(10) = \frac{500}{\sqrt{26}} \approx 98.1\text{ m}, \quad y = x/5$$

Continuing the straight climb indefinitely is unrealistic.

Further Example

Solution 34

A skater follows $x = 8\cos 20t,\ y = 12\sin(10t - \pi/3)$, where t is in minutes. Find the initial point, the self-intersection, the y-axis crossings and the time for a full circuit.



Initial point

$(x(0), y(0)) = (8, -6\sqrt{3})$

Crossings

The crossing is $(-4, 0)$.

On the y-axis, $y = \pm 3(\sqrt{6} + \sqrt{2})$ or $\pm 3(\sqrt{6} - \sqrt{2})$.

Circuit

$T = \pi/5\text{ min} = 12\pi\text{ s} \approx 37.7\text{ s}$

Exercise 8E Page 218-220 — Solution 35

1. A boat crosses a 75 m wide river, with eastward distance $x = 0.9t$ m and northward distance $y = -3.2t$ m, after t seconds.

a. Find the crossing time.

b. Find the southward drift.

c. Show the path is straight.

d. Find its speed.

a

$t = 75/0.9 = 250/3 \approx 83.3$ s

b

$y = -800/3 \Rightarrow 267$ m south (3 s.f.)

c

$$y = -\frac{32}{9}x$$

d

$v = \sqrt{0.9^2 + 3.2^2} \approx 3.32\,\text{m}\,\text{s}^{-1}$

Exercise 8E Page 218-220 — Solution 36

2. A descending plane is modelled by $x = 80t,\ y = 3000 - 9.1t$, with distances in metres and t in minutes.

a. Find its initial height.

b. Discuss the stated domain $0 \le t \le 329$.

c. Find the horizontal distance at landing.

a

$y(0) = 3000$ m

b

$y = 0 \Rightarrow t = 3000/9.1 \approx 329.67$

The stated upper bound 329 stops just before landing (height 6.1 m). The exact landing domain ends at $30000/91$; beyond that the model predicts negative height.

c

$x = 80(30000/91) \approx 26400$ m (3 s.f.)

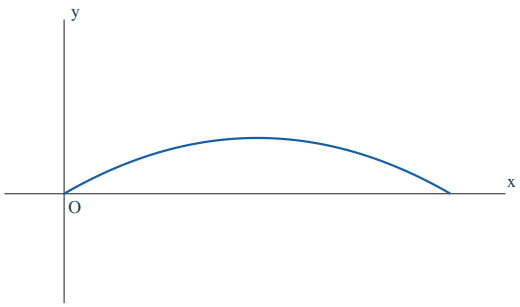
Exercise 8E Page 218-220 — Solution 37

3. A ball is kicked from the ground at $20\,\text{m s}^{-1}$, angle 30°. Its path is $x = 10\sqrt{3}t,\ y = 10t - 4.9t^2$.

a. Find its horizontal range.

b. Find when it is descending between heights 1.5 m and 2.5 m.

c. Find the nearest point where a player can head it in that interval.



a $t = 100/49 \Rightarrow x = 1000\sqrt{3}/49 \approx 35.3$ m

b
$$t_h = \frac{10 + \sqrt{100 - 19.6h}}{9.8}$$
Use the larger root for descent: $t_{2.5} \le t \le t_{1.5}$, approximately $1.75 \le t \le 1.88$ s.

c
$x_{\min} = 10\sqrt{3}\,t_{2.5} \approx 30.3$ m

Exercise 8E Page 218-220 — Solution 38

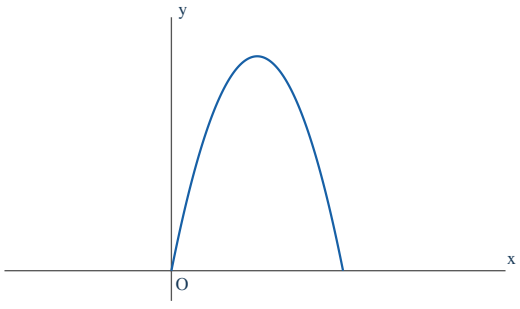
4. A dolphin leaps along $x = 2t,\ y = 10t - 4.9t^2$, in metres and seconds.

a. Find the jump duration.

b. Find its horizontal distance.

c. Show the path is quadratic.

d. Find its maximum height.



a $t(10 - 4.9t) = 0 \Rightarrow T = 100/49$ s

b $x = 200/49$ m

c
$$t = x/2 \Rightarrow y = 5x - \frac{49}{40}x^2$$

d
$t = 50/49, \quad y_{\max} = 250/49$ m

Exercise 8E Page 218-220 — Solution 39

5. A Ferris-wheel car has $x = 12\sin t,\ y = 12 - 12\cos t$, distances in metres and t in minutes.

a. Show its path is a circle of radius 12 m.

b. Find its maximum height.

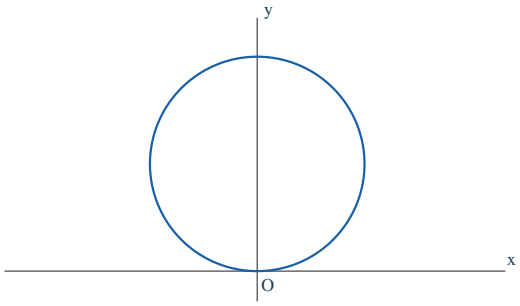
c. Find the period and average speed.

a $x^2 + (y-12)^2 = 144$

b $y_{\max} = 24$ m

c
$T = 2\pi$ min

$v = \dfrac{24\pi}{2\pi} = 12\,\text{m}\,\text{min}^{-1}$

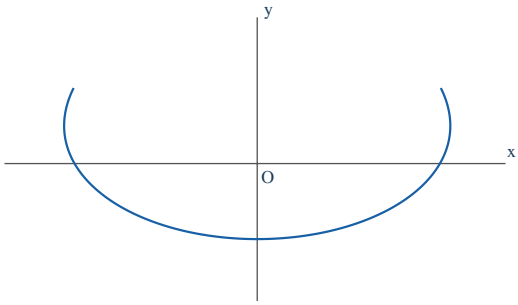


Exercise 8E Page 218-220 Q6 — Solution 40

6. A bowl has cross-section $x = t - 4\sin t,\ y = 1 - 2\cos t,\ -\pi/2 \le t \le \pi/2$.

a. Find the width of the opening.

b. Given that the deepest point is on the y-axis, find the depth.



a
$x(\pm\pi/2) = \pm(\pi/2 - 4)$

Width $= 8 - \pi \approx 4.86$

b
$y_{\text{rim}} = 1, \quad y(0) = -1, \quad \text{depth} = 2$

Explore an implicit curve

$\sin(\sin x + \cos y) = \cos(\sin(xy) + \cos x)$.

Not every curve has a convenient explicit or parametric representation. A contour plot draws the zero set of the difference between the two sides.

